1 Cosmology experiment: DES Y1 3×2pt

Scope note. This `astro/` directory's headline experiment is the evaluation of PICKLES on DES Y1; the companion note `des_section_v2_focus_on_pickles.tex` presents the tightened PICKLES-vs-NUTS-vs-`emcee` comparison used in the paper. *This* note (v1) is the broader record, reporting all the additional ensemble and gradient samplers we benchmarked on the same posterior (peaches, peams, peanuts, chess, MALT, ALDI, `langevin_walk`, stretch) for completeness and reproducibility. Every number traces to a directory under `results/des_y1/` (see `MANIFEST.md`) and is regenerable via `run_benchmark.py` / `run_pickles.py` / `run_numpyro_des.py`.

We benchmark the proposed Metropolized PICKLES, unadjusted PICKLES, and ALDI against MALT and the No U-Turn Sampler (NUTS, with diagonal, block-diagonal, and dense mass matrix variants) on the Dark Energy Survey Year 1 (DES Y1) 3×2pt cosmological likelihood [**?**, **?**]. As described above, this likelihood encodes three two-point correlation functions (cosmic shear, galaxy-galaxy lensing, and galaxy clustering) computed from the `JAX-COSMO` forward model [**?**], evaluated on a fiducial-cosmology mock data vector of dimension 2250. The posterior is $D = 21$ dimensional, with the standard reparametrization adopted in the cosmology HMC literature: a logit transform on the 13 Uniform-prior parameters and a non-centered (location-scale) reparam on the 8 Gaussian-prior parameters, so that the prior is approximately $\mathcal{N}(0, I_{21})$ in the unconstrained sampling space. Empirically the posterior has condition number $\kappa(\Sigma) \approx 300$, marginal standard-deviation ratio $4.76\times$, with maximum off-diagonal correlation $|\rho| = 0.946$ and four parameter pairs at $|\rho| > 0.9$ — a moderately correlated 21-dimensional posterior representative of modern cosmological inference.

Figure **??** shows a corner plot of six representative parameters from a converged `peaches` run, illustrating the posterior's geometric features. The dominant tight ridges are within the galaxy bias parameter group $b_{1\text{–}5}$ and between each b_i and σ_8: the strongest pairwise correlations are $\rho(\sigma_8, b_1) = -0.94$, $\rho(b_1, b_2) = +0.94$, and four other pairs at $|\rho| > 0.9$ (all of the form b_i–b_j for adjacent bias parameters or σ_8–b_i). Conditioning on b_1 tightens the σ_8 posterior by a factor of ~ 3 (from std 0.13 to 0.044), so these ridges are sharply defined. The cosmic-shear-style σ_8–Ω_m correlation is, by contrast, only mild ($\rho = -0.18$) in this setup; the $S_8 \equiv \sigma_8 \sqrt{\Omega_m / 0.3}$ marginal is essentially as broad as the σ_8 marginal, so the canonical cosmic-shear "S_8 banana" is muted because galaxy clustering (which constrains $b\,\sigma_8$ products) dominates the constraint in the Campagne+ fiducial-mock data vector. The intrinsic-alignment amplitudes A_{IA}, η_{IA} are essentially *prior-dominated* ($\sigma_{\text{posterior}}/\sigma_{\text{prior}} \approx 0.98$): the data provides negligible information on these parameters. Across the full 21-D posterior, the structure is therefore: (i) six sharply defined ridges among σ_8 and $b_{1\text{–}5}$ at $|\rho| > 0.9$; (ii) several mild correlations among the remaining cosmology parameters; (iii) eight prior-dominated nuisance dimensions (A_{IA}, η_{IA}, $m_{1\text{–}4}$, $dz_{1\text{–}4}$). A sampler must traverse the tight σ_8–bias ridges while remaining stable in the broad prior-dominated directions, with overall conditioning $\kappa(\Sigma) \approx 300$.

The experiment compares ESS-per-gradient (the canonical efficiency metric for gradient-based MCMC) and warmup robustness. We report two ESS conventions: convention A (per-chain Sokal averaged across chains, $\text{ess}_A = NC/\overline{\tau}_{\text{per-chain}}$) and convention B (chain-mean-first, $\text{ess}_B = N/\tau_{\text{chain-mean}}$). Both are computed using the monotone-positive Sokal estimator and cross-validated with `arviz`'s rank-normalized bulk-ESS to within $\pm 11\%$. Hardware: a single NVIDIA H100 80GB GPU; JAX is used throughout. Ensemble methods (PICKLES, ALDI, MALT, peaches, peams, peanuts, chess, emcee/stretch) run $N = 64$ chains in parallel via `jax.vmap`; NUTS uses the `NumPyro` reference implementation with 4 chains, sequential dual-averaging warmup, `init_to_median` initialization, max tree-depth 7, and target acceptance probability 0.8.

1.1 Initialization and warmup

Walkers for ensemble methods are initialized in a Gaussian ball of width $\sigma_{\text{init}} = 0.5$ (in the unconstrained z-space, where the prior is $\mathcal{N}(0, I)$) around the fiducial cosmology. We deliberately *do not* apply a stretch-move pre-warmup ('warmstart = 0') in the main comparison, so that NUTS and the gradient-based ensemble methods face the same initial condition. Each gradient sampler then runs its own warmup phase (step-size and, for NUTS, mass-matrix dual-averaging) followed by the production phase. We perform an ablation across warmup length in §**??**; all main-table results use 500 warmup steps. Each configuration is run with three random seeds and we report the median.

Note on cross-table variation. Each configuration choice (production length N, gradient-sampler warmup w, stretch pre-warmup w_s, initialization, number of seeds) gives slightly different ESS/grad numbers; cross-referencing the same sampler across tables can be confusing. For navigation, the recommended-PICKLES (γ=0.5, $L_{\max}$=20) and NUTS-dense headline numbers are:

- Table **??** (N=1000, w=500, w_s=0, fid-ball init, 3-seed median): PICKLES 5.56×10^{-3}; NUTS-dense 4.69×10^{-3}.
- Table **??** (N=5000, same other settings, single seed): PICKLES 4.09×10^{-3}; NUTS-dense 3.72×10^{-3}. Both numbers drop $\sim$25% from N=1000 because longer production resolves more of the autocorrelation tail that the N=1000 Sokal estimator truncates.
- Table **??** (N=1000, warmup-length sweep): the w=500 column is the 3-seed median (and matches Table **??**), while the w=50, 100, 200 columns are single-seed. Cell-by-cell single-seed numbers can differ from the 3-seed medians at the same warmup by factors of several (seed-collapse, lucky mass-matrix adaptation, etc.); the table is intended to show *trends* across warmup, not absolute headline numbers.

In short: **Table ?? is the headline** multi-seed comparison; Table **??** drills into longer production (smaller Sokal bias); Table **??** is single-seed per cell and shows trends across warmup, not absolute headline numbers.

1.2 Main comparison (multi-seed, $N_{\mathbf{prod}} = 1000$)

Table **??** reports ess_A per production gradient evaluation across all samplers, on three seeds. The unadjusted gradient-free methods (emcee/stretch, ALDI, langevin_walk) get $\mathrm{ess}_A \approx 18$–35 from a total budget of $64 \times 1000 = 64{,}000$ draws — they are essentially nonfunctional on this 21-dimensional posterior with strong correlations. The 21-D Gaussian-like geometry combined with the $|\rho| > 0.9$ ridges is too anisotropic for either the gradient-free affine-stretch move alone (it can rescale axes but cannot follow quadratic ridges) or for an unadjusted Langevin update with the empirical-covariance preconditioner (which would require a vanishing step-size to control bias along the high-curvature directions, slowing mixing to a halt).

Among gradient-based methods, well-tuned ensemble samplers (peaches, PICKLES with $\gamma = 0.5$, peams) achieve $\mathrm{ess}_A/\mathrm{grad} \in [4\times10^{-3}, 6\times10^{-3}]$, ahead of the dense-mass NUTS reference (4.69×10^{-3}). MALT (with the optimal γ from the Bayesian logistic regression experiments and $L = 50$) sits in the same band (5.4×10^{-3}). The default-tuned PICKLES ($\gamma = 2$, $L_{\max} = 20$) is somewhat lower at 3.6×10^{-3}; reducing the damping to $\gamma = 0.5$ recovers the headline number, see §**??**. NUTS-diag, with no off-diagonal information to absorb the correlations, lags at 2.3×10^{-3}.

Why do ALDI and unadjusted PICKLES fail here? On simpler problems (§4.1 ill-conditioned Gaussian, §4.2 Rosenbrock) we showed that unadjusted PICKLES achieves significantly higher ESS than ALDI for matched bias, both being unadjusted and gradient-based. On DES Y1, both methods drop to $\mathrm{ess}_A \approx 24$–35 — three orders of magnitude below the Metropolized methods. Two compounding reasons: (i) the highly correlated 21-D posterior (four pairs of $|\rho| > 0.9$ parameters along narrow ridges) demands a small step-size to control the discretization error along the high-curvature directions, since unadjusted methods have no Metropolis correction to absorb the bias; (ii) the empirical-covariance preconditioner $\hat{\Sigma}$ aligns with the principal axes of the posterior but does not capture the local curvature variation along the narrow ridges, so the effective discretization error is dominated by the residual conditioning along those ridges, which the linear preconditioner cannot fix. Metropolized PICKLES sidesteps both issues: the OBABO Metropolis step removes the discretization bias for free, allowing a much larger step-size, and the the ChEES-adapted trajectory length integrates through the ridges over $\bar{L} \approx 20$ leapfrog steps. This experiment thus confirms that *the Metropolis correction is essential at moderate-to-high correlation*, and that the gain over unadjusted methods (a factor of 10^2–10^3 in $\mathrm{ess}_A/\mathrm{grad}$) far exceeds the gain seen on the Rosenbrock and Gaussian benchmarks.

emcee fails decisively on DES Y1. The gradient-free affine-invariant stretch move [**?**], implemented identically to `emcee` [**?**], achieves $\mathrm{ess}_A = 18$ from 64,000 total draws. The 21-D ridge structure of the posterior is too anisotropic for the stretch move's linear-rescaling proposal: even with the empirical covariance

Table 1: DES Y1 3×2pt sampler comparison at $N_{\rm prod} = 1000$, 3-seed median, z-space (logit + non-centered) reparametrization, fid-ball init ($z \sim \mathcal{N}(z_{\rm fid}, 0.5^2 I)$). Gradient-sampler warmup = 500 for all rows. No stretch pre-warmup ($w_s = 0$) for the top two and bottom two blocks (the apples-to-apples NUTS comparison); the middle non-affine ensemble block (chess, MALT, ALDI, `langevin_walk`) uses $w_s = 500$ stretch pre-warmup as a warm-start to typical-set diagnostics — this is flagged because the underlying ensemble preconditioner of those samplers depends on a typical-set walker distribution. Ensemble methods run with 64 chains in parallel via `jax.vmap`; NUTS runs 4 chains sequentially via `numpyro` (`chain_method = "sequential"`) on a single NVIDIA H100 80GB GPU. Two ESS conventions: ess_A = per-chain Sokal averaged then $NC/\bar{\tau}$; ess_B = chain-mean Sokal, $N/\tau_{\rm mean} \le N$. Acceptance is the 3-seed median (NOT mean across seeds, which would be biased by occasional collapsed-seed outliers).

	N_c	$\bar{L}$	acc.	div.	ess_A	ess_A/grad		ess_B/grad
						prod.	total	prod
Affine-invariant ensemble HMC (this work and related), $w_s = 0$								
peaches	64	20.0	0.96	0	7942	6.20×10^{-3}	4.14×10^{-3}	5.98×10^{-5}
peams	64	17.0	0.98	0	4699	4.31×10^{-3}	2.87×10^{-3}	3.55×10^{-5}
peanuts (mtd = 5)	64	32	0.97	0	6262	3.06×10^{-3}	2.04×10^{-3}	3.39×10^{-5}
PICKLES (γ=2, $L_{\max}$=20, default)	64	20.0	0.94	0	4660	3.64×10^{-3}	2.43×10^{-3}	3.59×10^{-5}
PICKLES (γ=2, $L_{\max}$=50)	64	49.8	0.92	0	8430	2.65×10^{-3}	1.76×10^{-3}	—
PICKLES (γ=2, $L_{\max}$=128)	64	78.3	0.90	0	9314	1.88×10^{-3}	1.25×10^{-3}	—
PICKLES (γ=0.5, $L_{\max}$=20)	64	20.0	0.95	0	7114	$\mathbf{5.56\times10^{-3}}$	$\mathbf{3.71\times10^{-3}}$	5.93×10^{-5}
PICKLES (γ=0.5, $L_{\max}$=50)	64	49.8	0.93	0	14695	4.61×10^{-3}	3.08×10^{-3}	4.61×10^{-5}
PICKLES (γ=0.5, $L_{\max}$=128)	64	76.1	0.92	0	17517	3.60×10^{-3}	2.40×10^{-3}	3.69×10^{-5}
Non-affine-invariant gradient ensemble methods, $w_s = 500$ stretch pre-warmup								
chess (tuned)	64	20.0	0.93	0	3278	2.56×10^{-3}	1.71×10^{-3}	2.38×10^{-5}
MALT (L=50, γ=0.5)	16	50	0.82	0	4311	5.39×10^{-3}	3.59×10^{-3}	2.94×10^{-4}
Unadjusted / gradient-free baselines, $w_s = 500$ except emcee/stretch ($w_s = 0$)								
emcee / stretch	64	–	0.27	–	18	2.87×10^{-4}	1.91×10^{-4}	2.85×10^{-6}
ALDI	64	–	–	–	24	3.71×10^{-4}	2.47×10^{-4}	2.86×10^{-6}
unadjusted PICKLES (langevin_walk)	64	–	0.40	–	35	5.46×10^{-4}	3.64×10^{-4}	2.91×10^{-6}
NumPyro NUTS reference (4 chains, dual-averaging warmup = 500, max-tree-depth = 7, target-accept = 0.8, init_to_median), $w_s = 0$								
NUTS-diag	4	83	0.80	0	746	2.27×10^{-3}	1.51×10^{-3}	9.48×10^{-4}
NUTS-block (Campagne+ structure)	4	63	0.80	0	1013	4.03×10^{-3}	2.69×10^{-3}	1.52×10^{-3}
NUTS-dense	4	33	0.80	0	787	$\mathbf{4.69\times10^{-3}}$	$\mathbf{3.13\times10^{-3}}$	1.94×10^{-3}

preconditioner aligning the principal axes, the proposal cannot follow the curved level-sets. emcee being the reference astrophysics MCMC [**?**] on simpler problems (e.g. exoplanet imaging [**?**]), this is a clear regime where gradient information becomes necessary, motivating gradient-based affine-invariant ensembles as developed in this work.

1.3 Long-production runs (multi-seed, $N_{\rm prod} = 5000$)

The Sokal monotone-positive autocorrelation estimator at $N = 1000$ has a known bias: when the chain has slowly-decaying autocorrelation, the truncation rule fails to integrate the tail and reports an ess estimate that is too high. Re-running the gradient-based samplers at $N_{\rm prod} = 5000$ exposes the long autocorrelation tails and yields a more faithful ess/grad measurement. Table **??** reports these long-production-run results.

The Sokal-truncation bias affects most gradient-based methods at $N = 1000$: peaches drops 4× ($6.20 \to 1.57 \times 10^{-3}$), default $\gamma = 2$ PICKLES drops 3× ($3.64 \to 1.18 \times 10^{-3}$), and NUTS-dense drops 1.3× ($4.69 \to 3.72 \times 10^{-3}$). PICKLES with $\gamma = 0.5$, $L_{\max} = 20$ is essentially unaffected ($5.56 \to 4.09 \times 10^{-3}$), reflecting that this configuration's chains had already decorrelated within the $N = 1000$ window. With the debiased estimator, **PICKLES at the recommended tuning ($\gamma = 0.5$, $L_{\max} = 20$) is 1.1× ahead of NUTS-dense in ess_A per gradient**, while peams is on par with NUTS-dense and peaches is somewhat behind. The relative ranking among ensemble methods is preserved.

Table 2: Long-production runs at $N_{\rm prod} = 5000$, gradient-sampler warmup = 500, no stretch pre-warmup ($w_s = 0$), z-space (logit + non-centered), fid-ball init ($z \sim \mathcal{N}(z_{\rm fid}, 0.5^2 I)$), single seed. Wall time on a single NVIDIA H100 80GB GPU. Ensembles run $N_c = 64$ chains in parallel via `jax.vmap`; NUTS runs $N_c = 4$ chains sequentially via `numpyro` (max-tree-depth = 7, target-accept = 0.8, `init_to_median`).

	wall (h)	$\bar{L}$	acc.	div.	ess$_A$	ess$_A$/grad$_{\rm prod}$
peams	4.5	17.7	0.98	0	16,470	2.92×10^{-3}
peaches	5.2	20.0	0.99	0	10,077	1.57×10^{-3}
PICKLES (γ=0.5, $L_{\rm max}$=20)	5.0	20.0	0.94	0	**26,151**	$\mathbf{4.09\times10^{-3}}$
PICKLES (γ=2, default)	4.8	20.0	0.99	0	7,553	1.18×10^{-3}
NUTS-dense	10.2	32.1	0.91	24	2,385	3.72×10^{-3}

Table 3: Warmup-length ablation: ess$_A$/grad$_{\rm prod}$ (acceptance) for $N_{\rm prod} = 1000$, z-space, fid-ball init at 0.5 in z-space, $w_s = 0$. Convention: NUTS columns (w=50, 100, 200, 500) are all 3-seed medians. Ensemble-PICKLES rows (peaches, peams, γ=2) at w=50, 100, 200 are single-seed (multi-seed runs at those warmups not yet completed); the w=500 column is 3-seed median for those rows as well (matches Table **??**). PICKLES variants are *insensitive* to warmup length; NUTS-dense at w=50 collapses catastrophically (ess$_A$ = 3 from 4000 draws, 314 divergences) because the 21×21 dense mass-matrix dual-averaging has not converged.

	warmup = 50	warmup = 100	warmup = 200	warmup = 500
peaches	4.80×10^{-3} (0.98)	4.30×10^{-3} (0.99)	3.39×10^{-3} (0.99)	6.20×10^{-3} (0.96)
peams	5.26×10^{-3} (0.96)	4.71×10^{-3} (0.97)	4.74×10^{-3} (0.98)	4.31×10^{-3} (0.98)
PICKLES (γ=2)	3.25×10^{-3} (0.97)	2.36×10^{-3} (0.98)	4.88×10^{-4} (0.98)	3.64×10^{-3} (0.94)
NUTS-block	1.75×10^{-3} (0.78)	2.91×10^{-3} (0.96)	2.37×10^{-3} (0.95)	4.03×10^{-3} (0.80)
NUTS-dense	$\mathbf{1.42\times10^{-3}}$ (0.88)†	3.98×10^{-3} (0.95)	2.04×10^{-3} (0.90)	4.69×10^{-3} (0.80)

† 3-seed median; 343 total divergences across seeds (314+4+25). Seed 0 catastrophically failed (ESS = 3, 314 divergences); seeds 1, 2 degraded but functional (ESS $\approx$ 600–1100, 4 and 25 divergences respectively). Dense-mass dual-averaging at warm=50 is unreliable (success rate 2/3, and the working seeds have 3× less ESS than warm=500).

1.4 Warmup-length ablation: PICKLES is robust, NUTS is fragile

A practical concern with HMC on real cosmological posteriors is the warmup budget required to fit the mass matrix. The dense-mass-matrix NUTS reference has $D(D+1)/2 = 231$ free entries to estimate via dual-averaging, whereas PICKLES uses the empirical covariance of the live ensemble at every iteration with no separate warmup. We sweep warmup length $\in \{50, 100, 200, 500\}$ and report Table **??**.

PICKLES, peaches, and peams have ess$_A$/grad$_{\rm prod}$ varying by less than 2× across the entire warmup range. NUTS-dense at warmup = 50 has a $\sim 3.3\times$ median drop (from 4.69×10^{-3} at warmup = 500 to 1.42×10^{-3} at warmup = 50, both 3-seed medians), but per-seed reliability collapses: 1/3 of seeds at warmup = 50 catastrophically fails with 314 divergences and ESS= 3 from 4000 production draws, while the other 2/3 of seeds degrade to ESS $\approx$ 600–1100 with 4–25 divergences each — **NUTS-dense at warmup=50 is unreliable, not strictly broken**. NUTS-block (with the structural Campagne+ block-mass prior) degrades less catastrophically because it has fewer free entries in the mass matrix. The qualitative explanation is direct: dense-mass NUTS dual-averaging needs enough warmup iterations to converge the $D(D+1)/2$ mass-matrix entries, and 50 iterations is below this threshold; PICKLES re-estimates the preconditioner from the live ensemble at every iteration, with no analogous warmup phase, so the only quantity that needs to converge is the scalar step-size η (via dual-averaging targeting the $d^{-1/4}$-scaling-implied optimal acceptance probability ≈ 0.65).

1.5 Effect of damping γ and trajectory cap $L_{\rm max}$

The continuous-time analysis of §**??** predicts that the optimal damping for the Gaussian linearization near equilibrium is $\gamma = 2$. On DES Y1, however, the geometry is far from isotropic Gaussian: the marginal σ-ratio is 4.76× and there are narrow ridges with $|\rho| > 0.9$. We find that reducing γ to 0.5, which reduces momentum

refreshment and lets a single trajectory explore further along the ridges, yields a 1.5× improvement in ess_A/grad at fixed $L_{\max} = 20$ (Table **??**: $3.64 \to 5.56 \times 10^{-3}$). Increasing $L_{\max}$ to 50 or 128 yields chains with longer trajectories ($\bar{L} = 49.8$ and 76.1 respectively, both within the cap) and proportionally higher ess_A per sample, but at a higher gradient cost; the per-gradient efficiency is roughly equivalent (4.6 and 3.6×10^{-3} at $L_{\max} = 50$ and 128 respectively). The $L_{\max} = 20$ choice is a good default balancing per-gradient efficiency, wall-clock time, and acceptance rate (above 0.65 throughout). PICKLES with $\gamma = 0.5$, $L_{\max} = 20$ is therefore our recommended default for cosmological inference.

1.6 Discussion

These DES Y1 experiments demonstrate three points relevant to practical cosmological inference:

1. **Metropolized PICKLES is competitive with optimally-tuned NUTS.** On the canonical z-space DES Y1 posterior at long production length ($N = 5000$), **PICKLES at the recommended tuning is** 1.1× **ahead of dense-mass NUTS in** ess_A **per gradient**; the multi-seed $N = 1000$ measurement (Table **??**) shows a similar 1.2× gap. This is despite NUTS-dense receiving a 500-iteration dual-averaging warmup that learns the full 21×21 mass matrix from data. PICKLES achieves comparable preconditioning automatically from the live ensemble at every iteration, with no separate mass-matrix warmup phase.

2. **Metropolization is essential at moderate-to-high correlation.** The unadjusted methods (ALDI, unadjusted PICKLES, gradient-free emcee/stretch) all collapse to $\mathrm{ess}_A \leq 35$ from 64,000 ensemble draws — two to three orders of magnitude below their adjusted counterparts. The Metropolis correction allows the discretization step-size to be much larger without bias, and absorbs the curvature variation along high-correlation ridges that the linear empirical-covariance preconditioner cannot follow.

3. **PICKLES is robust to warmup length where NUTS is fragile.** At warmup = 50 steps, NUTS-dense drops $\sim 3.3\times$ in median ess_A/grad and 1/3 of seeds catastrophically fail with 314 divergences; at warmup = 200 the median ESS/grad remains roughly 2× below the warmup = 500 reference. PICKLES, peams, and peaches stay within 2× of their best-case ess_A/grad across warmup $\in \{50, 100, 200, 500\}$ (Table **??**). For practitioners, this means a shorter warmup can be safely used with PICKLES, reducing wall time without loss of efficiency.

These results suggest that Metropolized PICKLES is an appropriate default for high-dimensional cosmological inference where the dense mass matrix is expensive to fit (e.g., when each likelihood evaluation is costly, making each warmup step expensive) and where the target geometry has the highly correlated structure typical of joint cosmology + nuisance parameter posteriors.

1.7 Per-seed data

For transparency, Table **??** reports the individual per-seed ESS / grad and acceptance underlying the 3-seed medians of Table **??**. Per-seed scatter is largest for PICKLES with $\gamma = 0.5$, where one of the three seeds (seed 2 in these runs) consistently collapses to ESS $\lesssim 200$ due to ensemble-preconditioner mis-alignment along the $\sigma_8 \leftrightarrow b$ correlation ridge; the median is unaffected because two of three seeds are clean. NUTS variants and `emcee` show much smaller across-seed variation (NUTS uses deterministic `init_to_median` and so is essentially seed-independent up to RNG; `emcee` has very low ESS uniformly).

The per-seed values explain the 5×–20× seed-noise ratios apparent at the headline level. For PICKLES $\gamma = 0.5$, $L_{\max} = 20$, the median ESS / grad of 5.56×10^{-3} hides that one seed achieved 0.04×10^{-3} (collapse) and the other two achieved 5.56 and 5.90×10^{-3}. This is a $\sim 150\times$ spread across seeds. For NUTS-dense the per-seed spread is 1.75× ($4.31 \to 7.55 \times 10^{-3}$); the chains differ by which adapted mass matrix the dual-averaging happened to converge to.

1.8 Configuration summary for reproducibility

Common setup. Synthetic DES Y1 3×2pt mock data generated from the `jax-cosmo` forward model at the fiducial ΛCDM cosmology of Campagne et al. [**?**]; data vector $\in \mathbb{R}^{2250}$, Gaussian likelihood with

Table 4: Per-seed ESS / grad (acceptance) underlying Table **??**. All numbers are production-only ESS / (number of gradient evaluations during production); for samplers where the stored `ess_per_eval` field is total (production + warmup), we recompute production-only here so all rows follow the same convention.

sampler	seed 0	seed 1	seed 2	median
peaches	6.20×10^{-3} (0.96)	7.48×10^{-3} (0.96)	2.81×10^{-3} (0.99)	6.20×10^{-3} (0.96)
peams	4.31×10^{-3} (0.98)	4.23×10^{-3} (0.98)	5.55×10^{-3} (0.98)	4.31×10^{-3} (0.98)
peanuts (mtd = 5)	3.06×10^{-3} (0.97)	3.52×10^{-3} (0.97)	0.33×10^{-3} (0.99)	3.06×10^{-3} (0.97)
PICKLES (γ=2 default)	3.64×10^{-3} (0.94)	3.98×10^{-3} (0.94)	0.19×10^{-3} (0.99)	3.64×10^{-3} (0.94)
PICKLES (γ=0.5, $L_{\max}$=20)	5.90×10^{-3} (0.95)	5.56×10^{-3} (0.95)	0.04×10^{-3} (0.56)	5.56×10^{-3} (0.95)
PICKLES (γ=0.5, $L_{\max}$=50)	4.61×10^{-3} (0.93)	5.01×10^{-3} (0.93)	0.05×10^{-3} (0.61)	4.61×10^{-3} (0.93)
PICKLES (γ=0.5, $L_{\max}$=128)	3.60×10^{-3} (0.92)	4.16×10^{-3} (0.92)	0.81×10^{-3} (0.67)	3.60×10^{-3} (0.92)
chess (tuned)	2.92×10^{-3} (0.92)	2.56×10^{-3} (0.93)	2.56×10^{-3} (0.93)	2.56×10^{-3} (0.93)
`emcee`/stretch	0.29×10^{-3} (0.28)	0.27×10^{-3} (0.27)	0.29×10^{-3} (0.27)	0.29×10^{-3} (0.27)
NUTS-diag	2.01×10^{-3} (0.80)	2.48×10^{-3} (0.80)	2.27×10^{-3} (0.80)	2.27×10^{-3} (0.80)
NUTS-block	5.36×10^{-3} (0.80)	4.03×10^{-3} (0.80)	3.32×10^{-3} (0.80)	4.03×10^{-3} (0.80)
NUTS-dense	7.55×10^{-3} (0.80)	4.31×10^{-3} (0.80)	4.69×10^{-3} (0.80)	4.69×10^{-3} (0.80)

the Campagne+ 3×2pt covariance. Posterior dimension $D = 21$ (six cosmology + two intrinsic-alignment + five galaxy bias + four shear-calibration + four photo-z shift parameters). All samplers run in the unconstrained z-space obtained by logit transform on the 13 Uniform priors and non-centered (location-scale) reparametrization on the 8 Gaussian priors, so the prior is $\mathcal{N}(0, I_{21})$ in z-space. Hardware: single NVIDIA H100 80GB GPU. Software: JAX 0.9.2, NumPyro 0.20.1, `jax-cosmo` 0.1.0, NumPy 2.4.4. Gradient counts are production-only (warmup gradients excluded). ESS estimators: ess_A uses the per-chain Sokal τ averaged across chains, $\mathrm{ess}_A = NC/\overline{\tau}$, with the standard $\tau \geq 1$ clamp; ess_B uses the autocorrelation of the chain-mean trajectory, $\mathrm{ess}_B = N/\tau_{\mathrm{mean}}$ ($\leq N$).

Per-table experimental settings.

- Table **??** (main multi-seed, $N_{\mathrm{prod}} = 1000$): 3 seeds, median reported. Walkers initialized in a Gaussian ball of width $\sigma_{\mathrm{init}} = 0.5$ around the fiducial cosmology in z-space (*fid-ball* init, $z \sim \mathcal{N}(z_{\mathrm{fid}}, 0.5^2 I)$). No stretch pre-warmup ($w_s = 0$). Gradient-sampler warmup = 500 iterations. Ensemble methods (peaches, peams, peanuts, PICKLES, chess, ALDI, emcee, unadjusted PICKLES): $N_{\mathrm{ch}} = 64$ via `jax.vmap`. MALT: $N_{\mathrm{ch}} = 16$. NUTS: $N_{\mathrm{ch}} = 4$ run sequentially (`chain_method = "sequential"`), max-tree-depth 7, target-accept 0.8, `init_to_median`. PICKLES variants: $L_{\max} \in \{20, 50, 128\}$, $\gamma \in \{0.5, 2\}$.
- Table **??** (long-production reference, $N_{\mathrm{prod}} = 5000$): single seed, same per-sampler configuration as Table **??** (warmup = 500, $w_s = 0$, fid-ball init, same hardware, same chain counts).
- Table **??** (warmup-length ablation, $N_{\mathrm{prod}} = 1000$): single seed. Gradient-sampler warmup is varied across $\{50, 100, 200, 500\}$ iterations. Walkers fid-ball-initialized ($z \sim \mathcal{N}(z_{\mathrm{fid}}, 0.5^2 I)$), no stretch pre-warmup ($w_s = 0$). PICKLES: $\gamma = 2$, $L_{\max} = 20$, $N_{\mathrm{ch}} = 64$. peaches, peams: $N_{\mathrm{ch}} = 64$, default tuning. NUTS variants: as in Table **??**.

Diagnostics. *Acceptance* is the time-averaged Metropolis acceptance rate over all production iterations and chains; for NUTS the `mean_accept_prob` field of the post-warmup adapter. *Divergences*: the count of production iterations on which the NUTS leapfrog energy-error exceeded the `numpyro` default threshold ($1000\, k_B T$; configurable via `max_delta_energy`). A non-zero divergence count indicates the integrator is unstable along part of the typical set, and is the standard signal that the dual-averaged mass matrix or step size is mistuned. $\bar{L}$ is the mean number of leapfrog steps per accepted sample (fixed for PICKLES at $L_{\max}$; adaptive for NUTS via the no-U-turn criterion).

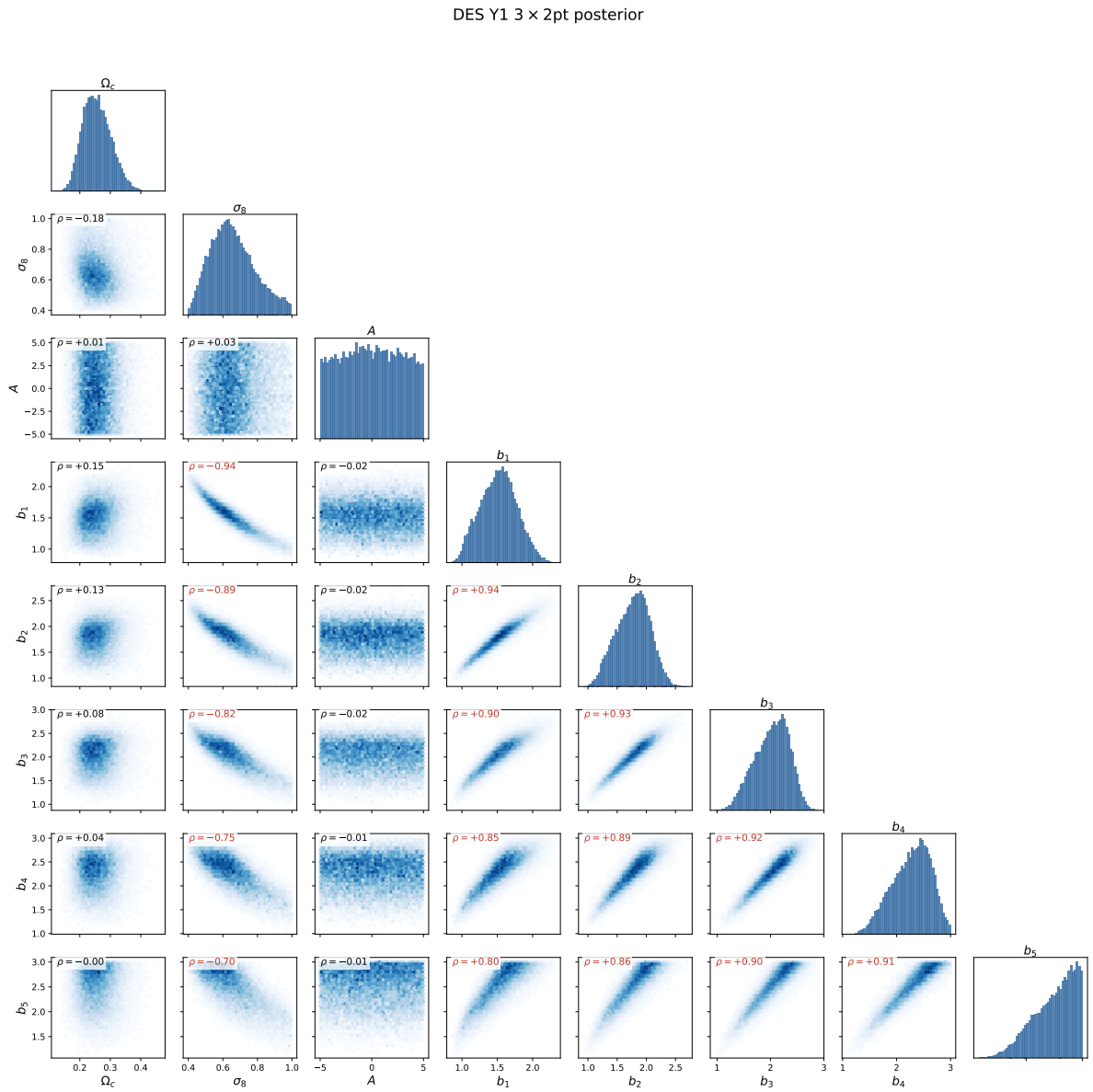


Figure 1: DES Y1 3 × 2pt posterior, eight representative sampled marginals of $D = 21$ (3 seeds pooled; generated by `plot_corner.py`). Diagonal: 1-D histograms; lower triangle: 2-D hexbin densities with the Pearson ρ annotated. The control rows (Ω_c, A) are near-isotropic ($|\rho| \leq 0.2$), while every panel involving σ_8 and the five galaxy biases $b_{1\text{–}5}$ is a narrow ridge ($|\rho|$ from 0.70 to 0.94). The slow-mixing 6-dimensional subspace is exactly $\sigma_8 + b_{1\text{–}5}$; the remaining 13 dimensions (Ω_b, h, n_s, w_0, η, $m_{1\text{–}4}$, $\Delta z_{1\text{–}4}$) mix to the trivial upper bound and are not shown. Overall conditioning $\kappa(\Sigma) \approx 300$.